

9.5-9.7

$$p^2 = \mathbf{p} \cdot \mathbf{p} = p_x^2 + p_y^2 + p_z^2 \text{ and } r = |\mathbf{r}| = (\mathbf{r} \cdot \mathbf{r})^{1/2} = (x^2 + y^2 + z^2)^{1/2}$$

Rotational symmetry

$$p^2 = \mathbf{p} \cdot \mathbf{p} = p_x^2 + p_y^2 + p_z^2 \text{ and } r = |\mathbf{r}| = (\mathbf{r} \cdot \mathbf{r})^{1/2} = (x^2 + y^2 + z^2)^{1/2}$$

Orbital angular momentum is $\hat{L}_z$

$$[\hat{H}, \hat{L}_z] = 0$$

$$\begin{aligned} [\hat{L}_z, \hat{p}_x] &= [\hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \hat{p}_x] = [\hat{x}\hat{p}_y, \hat{p}_x] \\ &= [\hat{x}, \hat{p}_x]\hat{p}_y = i\hbar\hat{p}_y \end{aligned} \quad (9.70a)$$

$$\begin{aligned} [\hat{L}_z, \hat{p}_y] &= [\hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \hat{p}_y] = -[\hat{y}\hat{p}_x, \hat{p}_y] \\ &= -[\hat{y}, \hat{p}_y]\hat{p}_x = -i\hbar\hat{p}_x \end{aligned} \quad (9.70b)$$

$$[\hat{L}_z, \hat{p}_z] = [\hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \hat{p}_z] = 0 \quad (9.70c)$$

$$[\hat{L}_x, \hat{L}_x] = i\hbar\hat{L}_y \quad [\hat{L}_x, \hat{L}_y] = -i\hbar\hat{L}_x \quad [\hat{L}_x, \hat{L}_z] = 0$$

$$[\hat{L}_z, \hat{x}] = i\hbar\hat{y}$$

$$[\hat{L}_z, \hat{y}] = -i\hbar\hat{x}$$

$$[\hat{L}_z, \hat{z}] = 0$$

This radial equation has the same form as the one-dimensional Schrödinger equation

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \langle x|E \rangle = E \langle x|E \rangle \quad (9.99)$$

but with an **effective potential energy**

$$V_{\text{eff}}(r) = \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \quad (9.100)$$

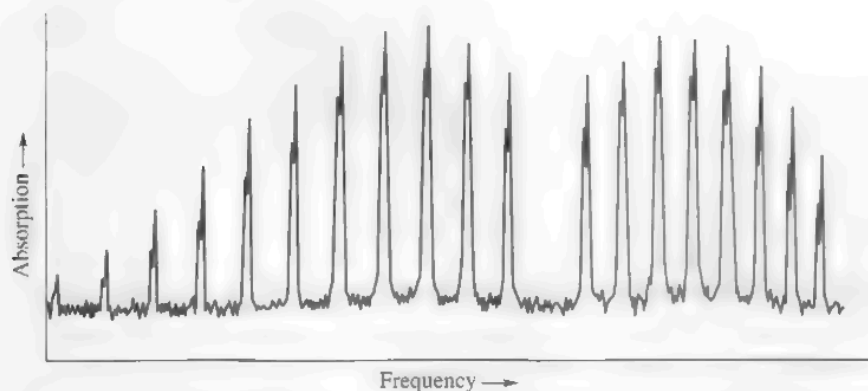


Figure 9.9 A vibrational-rotational absorption spectrum of HCl. Each peak is double because of the presence of two isotopes of chlorine in the gas— ^{35}Cl and the less abundant ^{37}Cl . Data are from M. Liu and W. Sly, Harvey Mudd College.